Statistics Lecture 4
Regression

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AALBORG UNIVERSITY
DENMARK

Connectivity

Schedule

Lecture	When
1. Introduction to statistics	March 12
2. Parameter estimation	March 24
3. Confidence intervals	March 26
4. Regression	**March 31**
5. Hypothesis testing pt. 1	April 7
6. Hypothesis testing pt. 2	April 9
Workshop: Wrap-up and exam problems	TBD

Outline

Linear regression

Least squares estimators

Residual analysis

Summary

linear function

$y = ax + b$

slope y-intercept

Linear regression

Where is life expectancy higher?

No binary distinction

Independent variable (x-axis):

- Life satisfaction index

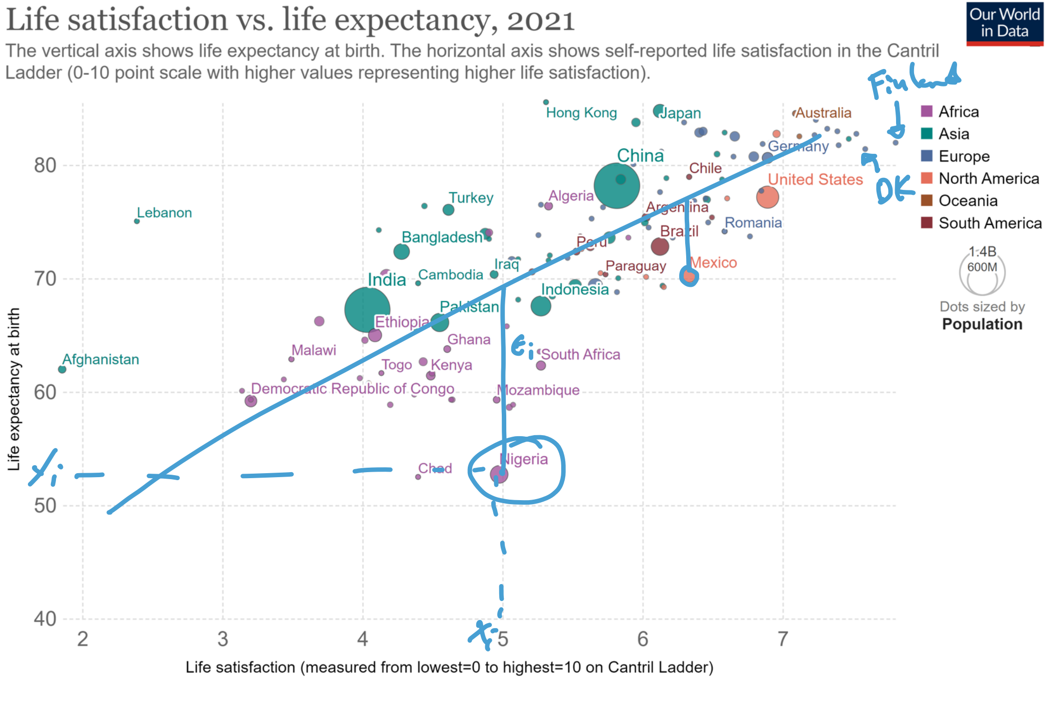
Dependent variable (y-axis):

- Life expectancy at birth

We start from scatter plot

Can we see a trend?

Can we use it for prediction?



Source: United Nations – Population Division (2022); World Happiness Report (2023) OurWorldInData.org/happiness-and-life-satisfaction • CC BY

Linear regression

We have data in the form $(X_1, Y_1), \ldots, (X_n, Y_n)$

And assume that the relationship between variables is **linear**

$$\boxed{Y_i = \beta_0 + \beta_1 X_i} + \underbrace{\epsilon_i}_{\text{noise}},$$

Where

- β_0 is the y-intercept
- β_1 is the slope
- ϵ_i is the error (also called the noise)
- $\mathbb{E}(\epsilon_i | X_i) = 0$
- $\mathrm{var}(\epsilon_i | X_i) = \sigma^2$

Simple linear regression

fixed

The parameters β_0 and β_1 are **unknown**: **we have to estimate them**

We end up with $\hat{\beta}_0$ and $\hat{\beta}_1$ so the **fitted line** is

RV

$$\hat{r}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

The **predicted values** are

$$\hat{Y}_i = \hat{r}(X_i)$$

And the **residuals** are

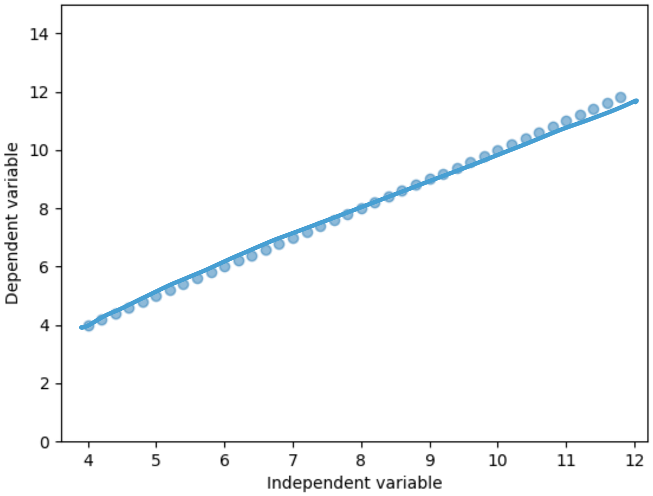
$$\hat{\epsilon}_i = Y_i - \hat{Y}_i = Y_i - \underbrace{\left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right)}_{Y_i}$$

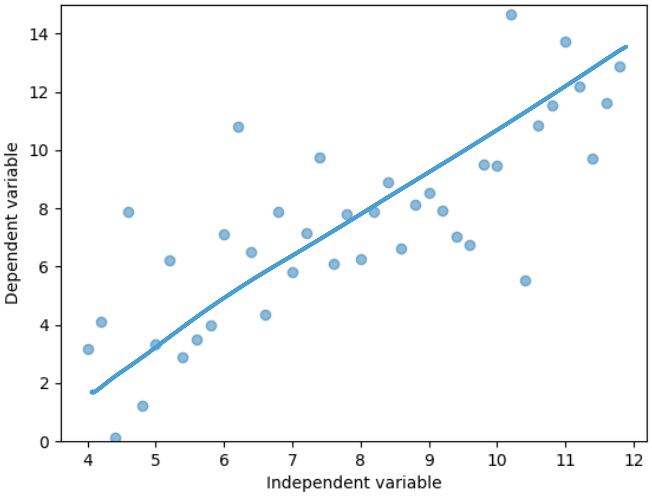
for X_i

> There is also multiple regression
>
> $$Y_i = \beta_0 + \beta_1 X_i^{(1)} + \cdots + \beta_k X_i^{(k)} + \epsilon_i$$

Intuitive explanation

We are recovering the underlying function after removing the noise





Least squares estimators

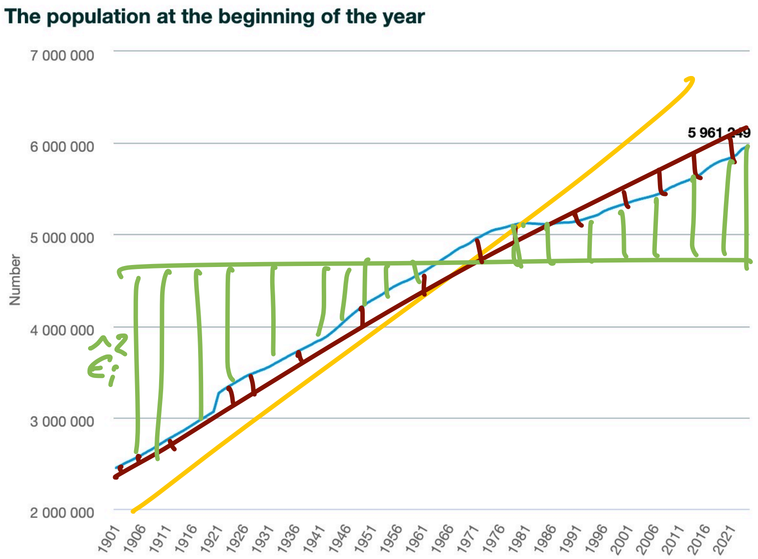
Determining the best estimate

In general, it is impossible to find a line that passes over all the points

What is the best fitted line?

$$\hat{r}(x) = \hat{\beta}_0 + \hat{\beta}_1 x$$

The one with the best estimators $\hat{\beta}_0$ and $\hat{\beta}_1$



Source: Statistics Denmark. https://www.dst.dk/en/Statistik/emner/borgere/befolkning/befolkningstal

Least squares regression

metric

Approach: Minimize the $\boxed{\text{Mean Square Error (MSE)}}$ for the residuals

$$\hat{\epsilon}_i = Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right)$$

Specifically, we minimize the Residual Sum of Squares (RSS)

$$\min_{\hat{\beta}_0,\hat{\beta}_1} \sum_{i=1}^{n} \hat{\epsilon}_i^{\,2} = \min_{\hat{\beta}_0,\hat{\beta}_1} \sum_{i=1}^{n} \left(Y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 X_i\right)\right)^2$$

sum squared

Finding the least squares estimates

Estimators

$$\hat{\beta}_1: \quad \frac{\partial \sum_{i=1}^{n} \left(Y_i - (\beta_0 + \beta_1 X_i)\right)^2}{\partial \beta_1} = 0$$

$$\hat{\beta}_0: \quad \frac{\partial \sum_{i=1}^{n} \left(Y_i - (\beta_0 + \beta_1 X_i)\right)^2}{\partial \beta_0} = 0$$

The least squares estimates

Estimators

$$\underline{\hat{\beta}_1} = \frac{\sum_{i=1}^{n}(X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}$$

slope

$$\underline{\hat{\beta}_0} = \bar{Y}_n - \hat{\beta}_1 \bar{X}_n$$

y-intercept

Unbiased estimator for σ^2

$$\hat{\sigma}^2 = \left(\underbrace{\frac{1}{n-2}}_{\text{dofs}}\right) \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

noise variance

RSS

Least squares under Gaussian noise

mean zero

Usually, we assume Gaussian noise:

$$\epsilon_i \sim N(0, \sigma^2)$$

variance

Therefore, for a given X_i we have

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

Least squares regression makes sense with Gaussian noise

Not so much sense if the noise is heavy-tailed. **Why?**

The connection with MLE

variance

With n i.i.d. RVs $Y_i \sim N(\overbrace{\beta_0 + \beta_1 X_i}^{\mu_i}, \sigma^2)$, the likelihood function is

value

mean

$$\mathcal{L}_n(\beta_0, \beta_1) = \prod_{i=1}^{n} \underbrace{f(Y_i; X_i, \beta_0, \beta_1, \sigma^2)}_{\text{PDF}} = \prod_{i=1}^{n} \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_i - \overbrace{\beta_0 - \beta_1 X_i}^{\text{mean}})^2}{\underbrace{2\sigma^2}_{\text{var}}}\right)}_{\text{PDF of } N(\beta_0+\beta_1 X_i, \sigma^2)}$$

The log-likelihood function is

$$\ell_n(\beta_0, \beta_1) = \log \mathcal{L}_n(\beta_0, \beta_1) = \boxed{\overset{\text{fixed}}{n \log \frac{1}{\sqrt{2\pi\sigma^2}}}} \boxed{- \frac{1}{2\sigma^2} \sum_{i=1}^{n} \underbrace{(Y_i - \beta_0 - \beta_1 X_i)^2}_{e_i^2}}$$

The parameters that **maximize the likelihood function** are the same ones that minimize the summation and the terms inside the summation are the residuals.

The least squares estimates are the same as the for the MLE

Distribution of $\hat{\beta}_1$

Continue at 9:12

Recall that $\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}$

Sample variance $S_X^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$

sort of...

biased version

True mean

noise var

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_X^2 n}\right)$$

var

$S_X^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$

larger sample size = smaller var

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}\right)$$

Distribution of $\hat{\beta}_0$

Recall that $\hat{\beta}_0 = \bar{Y}_n - \hat{\beta}_1 \bar{X}_n = \bar{Y}_n - \underbrace{\left(\frac{\sum_{i=1}^{n}(X_i - \bar{X}_n)(Y_i - \bar{Y}_n)}{\sum_{i=1}^{n}(X_i - \bar{X}_n)^2}\right)}_{\hat{\beta}_1} \bar{X}_n$

And $Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$

Sample variance $S_X^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$

biased est.

true mean

noise var

$$\hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2}{S_X^2 n}\left(\frac{\sum_{i=1}^{n} X_i^2}{n}\right)\right)$$

sum of vals. in x-axis

n goes away

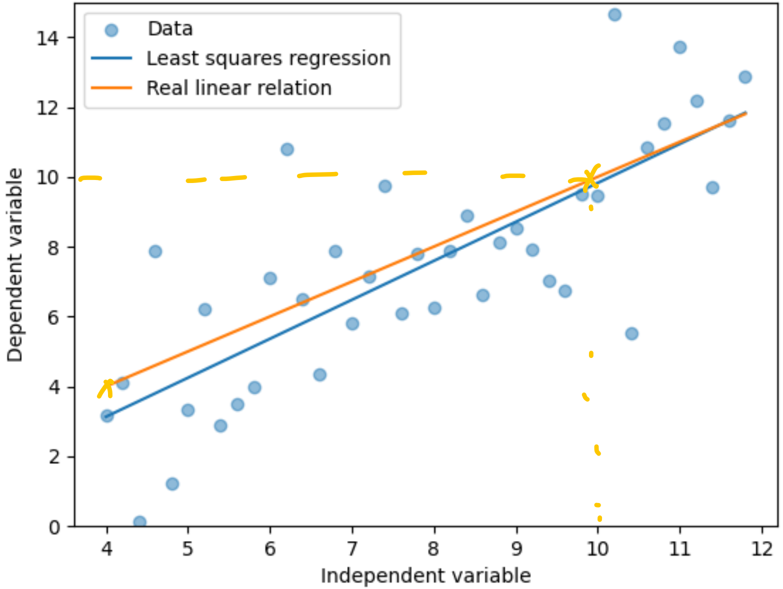
Example

Underlying model

$$Y_i = X_i + \epsilon_i = 0 + 1X_i + \epsilon_i$$

Estimated model with least squares

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i = -1.332 + 1.115X_i$$



Example

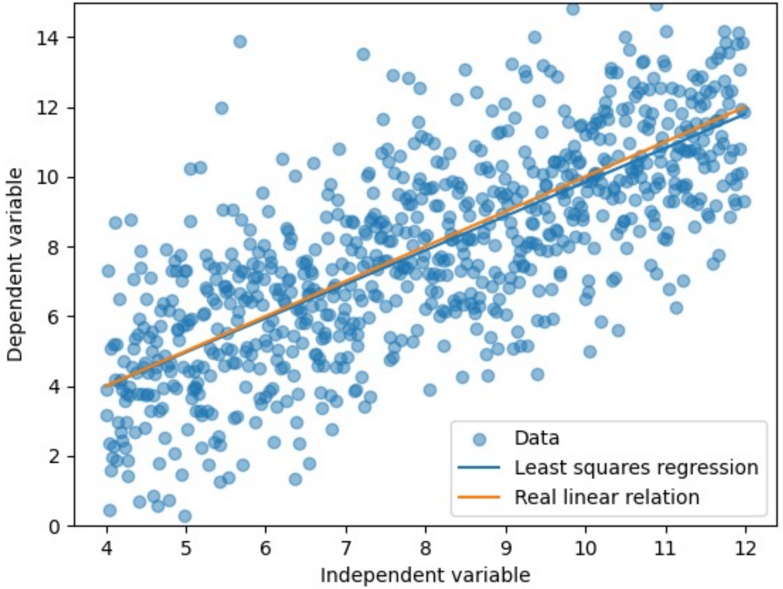
Underlying model

$$Y_i = X_i + \epsilon_i$$

Estimated model with least squares

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i = \underline{0.094} + \underline{0.974} X_i$$

should be 0 should be 1



Using the model: Mean response

We "train" the regression model with n data points and a new one x_* appears

1. The estimate of Y_* is also called the **mean response**

$$Y_* = \hat{\beta}_0 + \hat{\beta}_1 x_*$$

$Y_i = \beta_0 + \beta_1 X_i + \epsilon$

noise

2. What is the **confidence interval** for the mean response with a new value x_*?

$$C_{1-\alpha} = \boxed{\hat{\beta}_0 + \hat{\beta}_1 x_*} \pm T_{n-2,\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_* - \bar{X}_n)^2}{\sum_{i=1}^{n} (X_i - \bar{X}_n)^2} \right)}$$

mean quantile of t-dist.

Example: What would be the **mean price** for a 100 m² house in Copenhagen?

Using the model: Predicting a new response

We "train" the regression model with n data points and a new one x_* appears

1. The use the **mean response**

$$Y_* = \hat{\beta}_0 + \hat{\beta}_1 x_*$$

2. We can give a **prediction interval** for a new response

$$\hat{\beta}_0 + \hat{\beta}_1 x_* \pm T_{n-2,\alpha/2} \sqrt{\hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_* - \bar{X}_n)^2}{\sum_{i=1}^{n} (X_i - \bar{X}_n)^2} \right)}$$

wider than for the mean response

Example: What would be the **price** for a 100 m² house in Copenhagen?

Example: Predicting the population in Denmark

Data for regression from 1901 to 2024

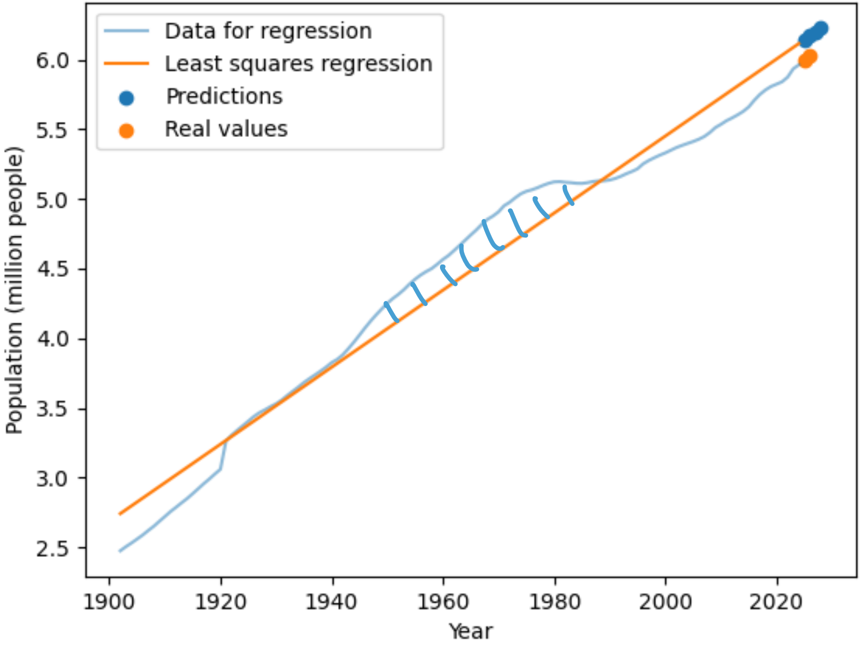
Regression:

$\hat{\beta}_0 = -49.803$ and $\hat{\beta}_1 = 0.02762731$

Prediction: $Y_* = \hat{\beta}_0 + \hat{\beta}_1 x_*$

$x_* = \{2025, 2026, 2027, 2028\}$

Year	Predictions	Real values
2025	6.087	5.99
2026	6.11	6.02
2027	6.197	----
2028	6.23	----



Data source: Statistics Denmark. https://www.dst.dk/en/Statistik/emner/borgere/befolkning/befolkningstal

Example: Predicting the population in Denmark

Data for regression from 1901 to 2024

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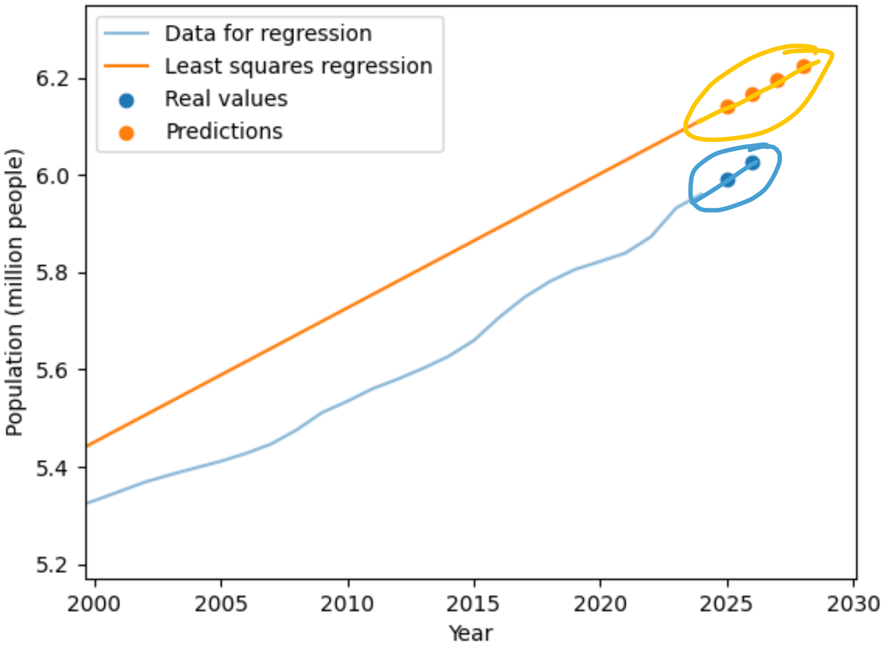

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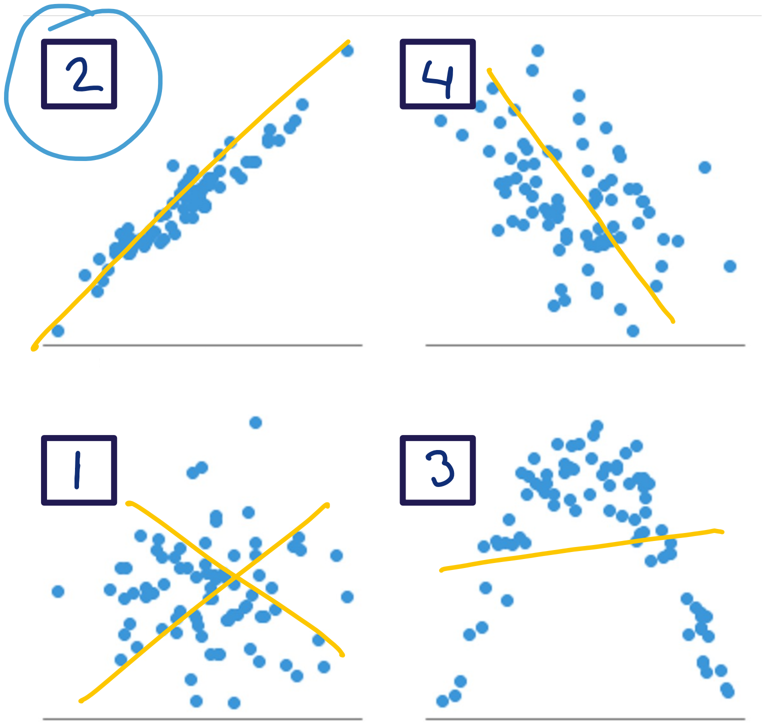
Residual analysis

When to use linear regression



1. No observable trend
2. Good prediction with linear regression
3. Not linear, leading to wrong predictions
4. Good average prediction but high variance

Good scenario for linear regression

Standardized residuals

$$Z_i = \frac{(Y_i - \beta_0 - \beta_1 X_i)}{\hat{\sigma}}$$

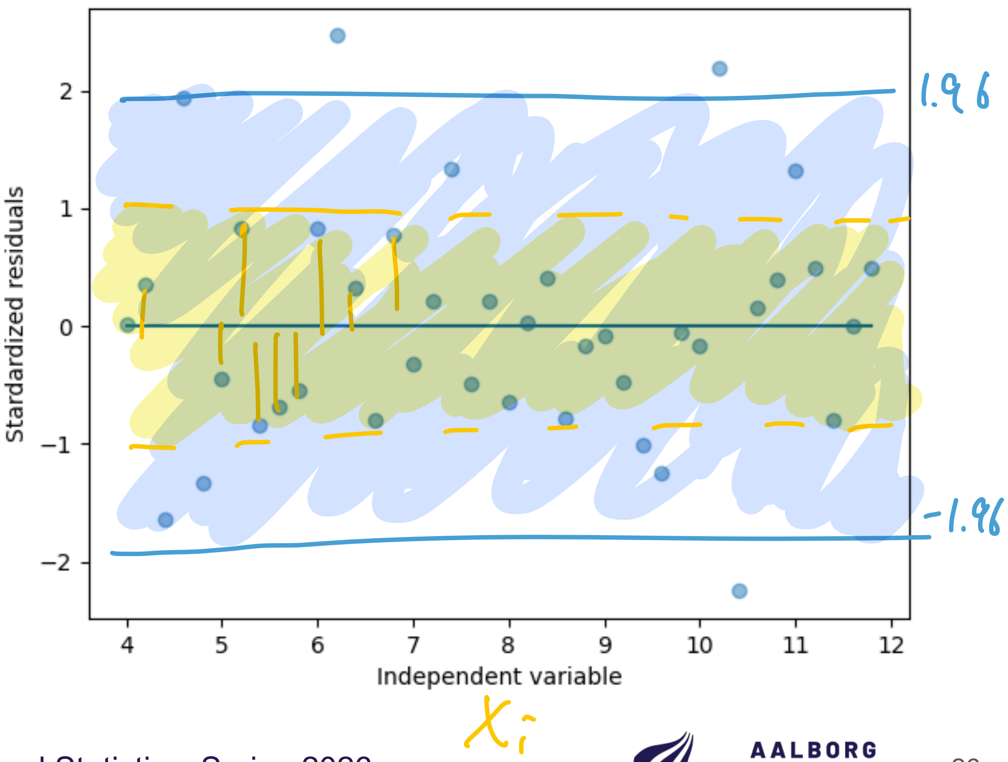
Measured in std. deviation units

$$\hat{\sigma}^2 = \left(\frac{1}{n-2}\right)\sum_{i=1}^{n} \hat{\epsilon}_i^2$$

Should be no correlation with x-axis

68% between -1 and 1

95% between -1.96 and 1.96



Coefficient of determination

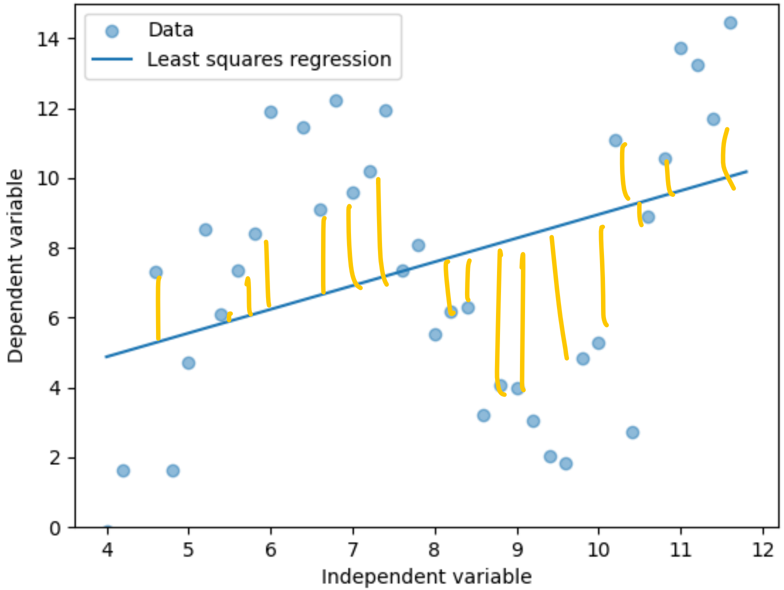
Proportion of the variation in the dependent variable that is predictable

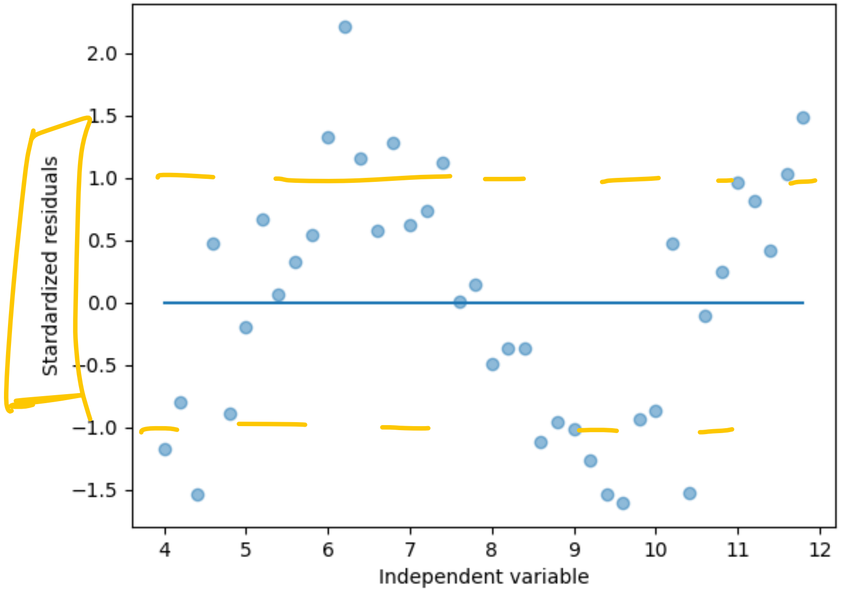
SSR

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=1}^{n} \hat{\epsilon}_i^2}{\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2} = 1 - \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2}$$

The best possible result is 1

Bad scenario for linear regression: model is not linear



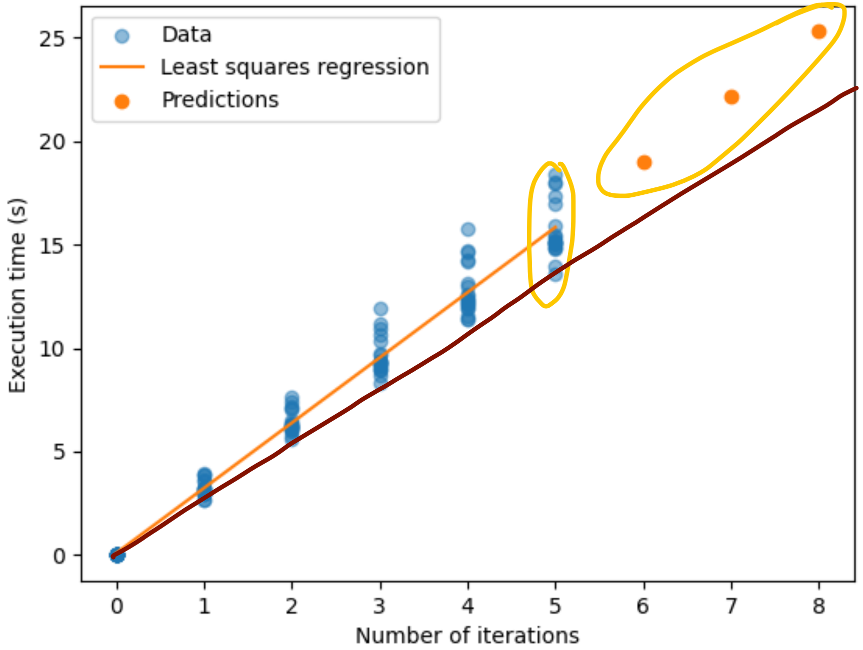


Not so good for linear regression

✓ The data follows a linear trend

Problem:

Large errors occur more for large values of x



Transforming the data

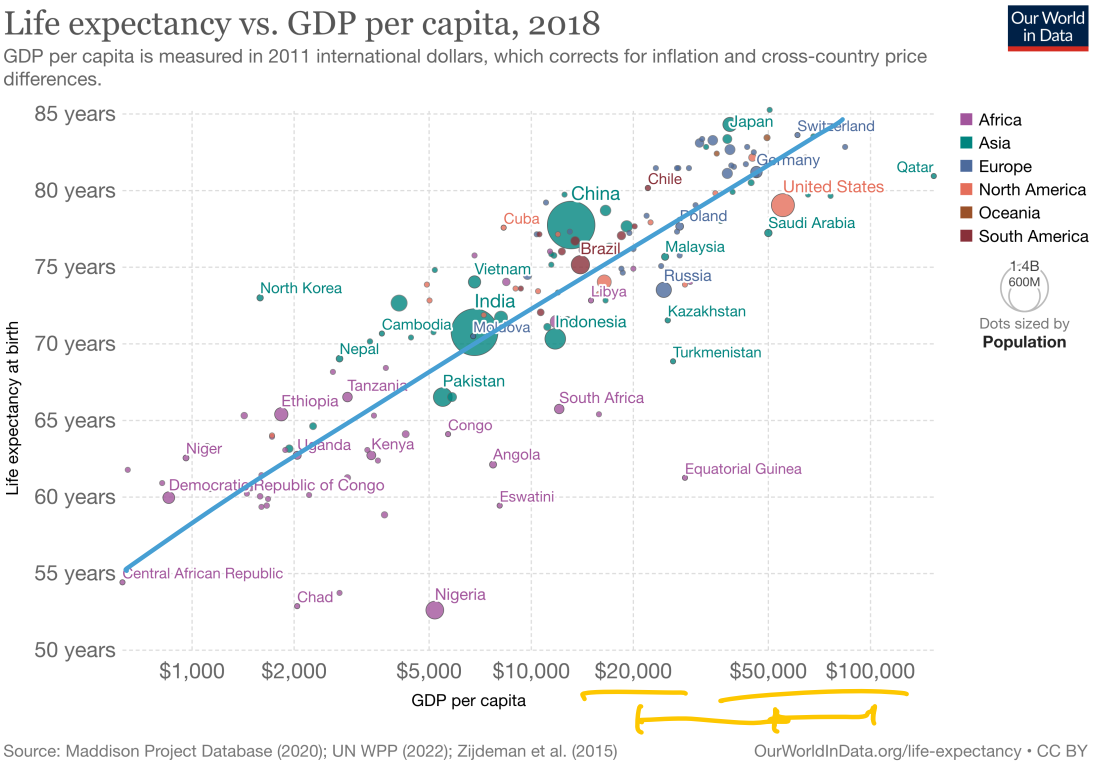
X-axis is not linear!

But it can be transformed

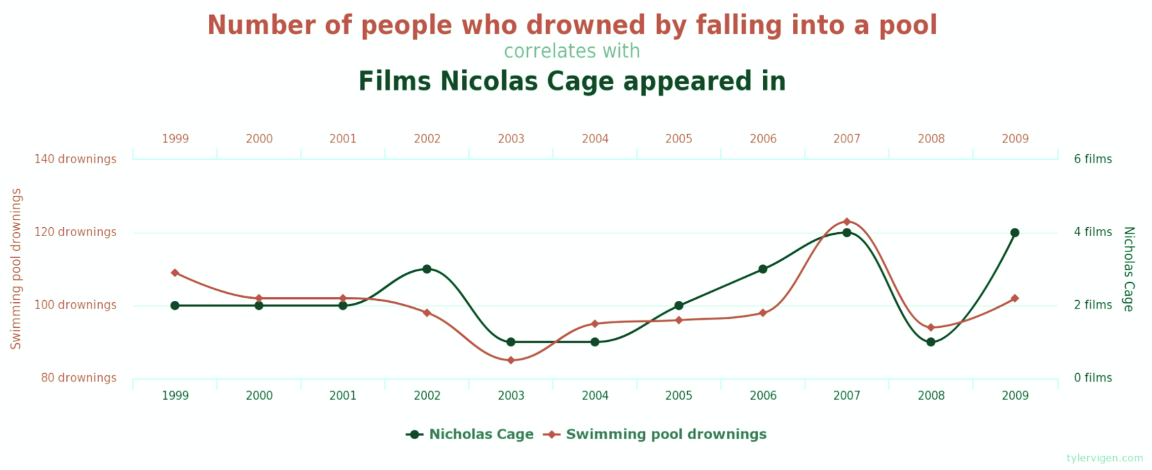
$\log(x_i)$

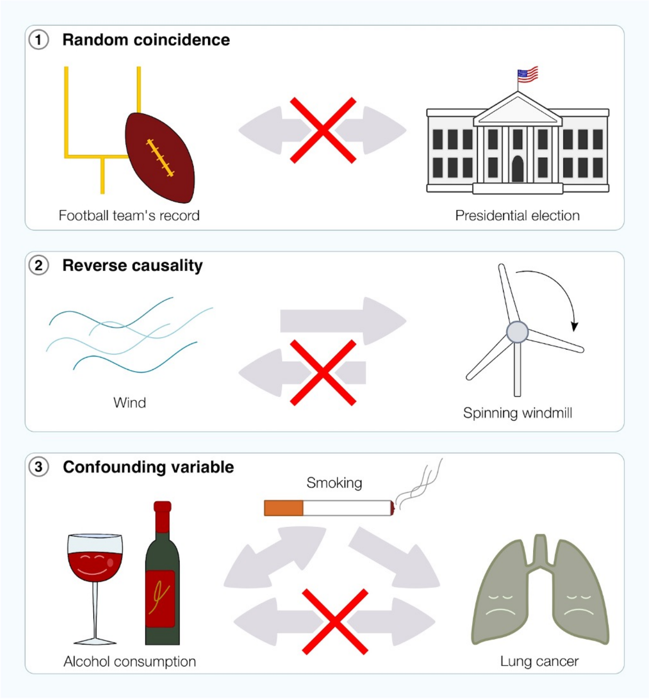
No linear relationship

Careful how the conclusion is stated

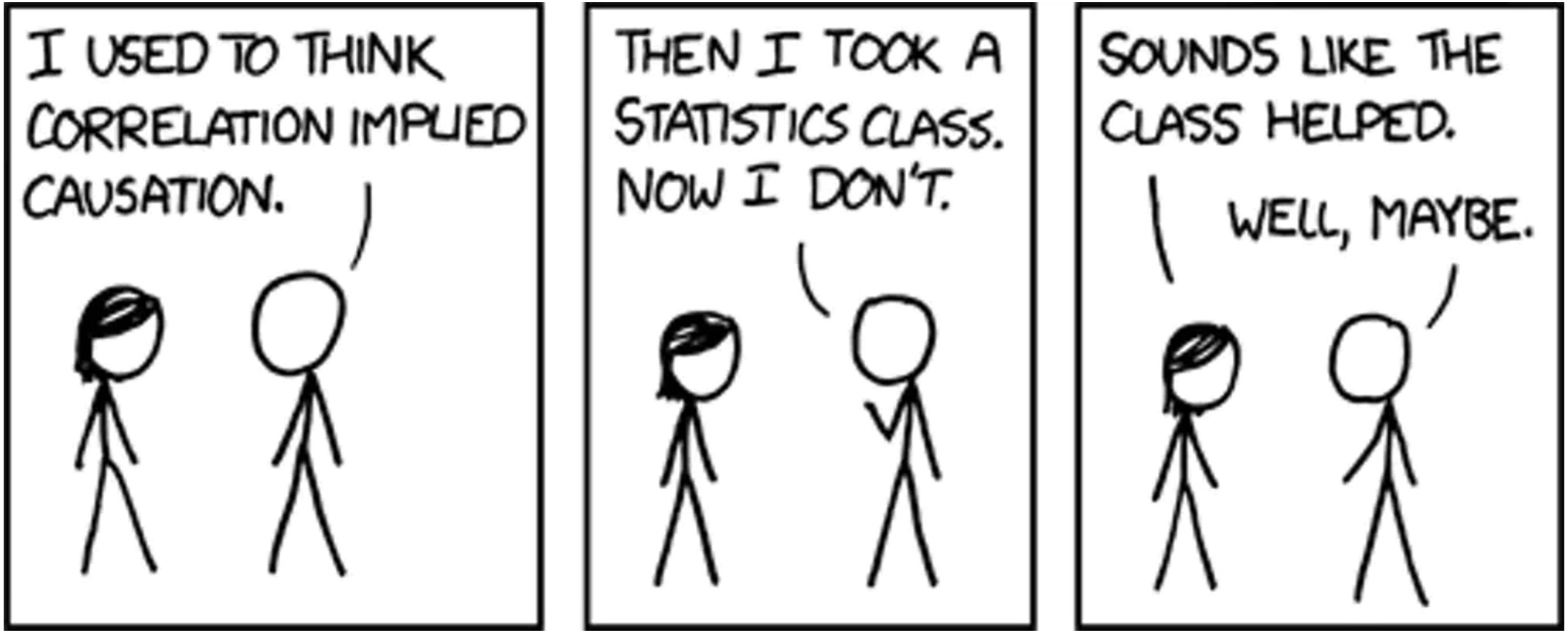


Correlation does not imply causation





Words of wisdom



Summary

Summary

Regression helps us identify correlation

Linear regression assumes the underlying model is linear

Besides fitting, we can perform:

- Hypothesis testing
- Prediction of the mean
- Prediction of a single new value

Residuals help us determine goodness of fit: is the model linear?

As initial test, try regression with a linear model with known parameters

That’s all for the lectures